

Priyanshu Paul

Guwahati, Assam

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EDUCATION

VIT Bhopal University	Sept'23 – May'27
Bachelor of Technology - Computer Science Engineering (AI&ML)	CGPA: 9.10/10
Faculty Higher Secondary School, Guwahati, Assam	Mar'22
Higher Secondary (Class: XII) - Science (PCM[B])	Grade: 95.8%
Faculty Higher Secondary School, Guwahati, Assam	Mar'20
High School (Class: X) - CBSE	Grade: 96.4%

SKILLS

Languages:	C++, Java, Python
Frontend:	HTML, CSS, JavaScript, React.Js, TailwindCSS, Next.Js
Backend and Databases:	PostgreSQL, Prisma, MySQL, Node.Js, REST APIs
Tools:	Git/GitHub, Postman, Vercel, AWS, Figma
Generative AI:	LangChain, RAG, Gemini/Claude APIs, HuggingFace, Vector DBs

PROJECTS

- CodePilot - AI-Powered Cloud-Based Code Editor [Demo](#) / [GitHub](#)
 - Architected a browser-based IDE with an autonomous **LLM agent** (Inngest AgentKit) enabling multi-step file create/edit/delete via natural language, backed by durable job execution and real-time UI streaming via **Convex** reactive database.
 - Engineered **ghost-text inline completions** (CodeMirror 6), **Cmd+K selection editing**, and full **WebContainers** in-browser Node.js execution, while integrating **Sentry** for LLM token/latency monitoring and **Clerk** for SaaS subscription billing, replicating core Cursor/VS Code capabilities and shipping a production-grade, monetizable developer tool.
 - Technologies:** Next.js, TypeScript, Convex, Inngest AgentKit, CodeMirror 6, WebContainers, Clerk, Sentry.
- Plabon - Corrective-RAG Flood & Disaster Relief Assistant [Demo](#) / [GitHub](#)
 - Built a **Corrective-RAG (C-RAG)** assistant to address Assam's annual flood crisis, where dense government SOPs are inaccessible in emergencies and generic LLMs risk hallucinating life-critical guidance.
 - Engineered a **corrective grading pipeline** (Gemini 2.5 Flash-Lite) that labels retrieved chunks as relevant/ambiguous/irrelevant across 3 branching generation paths, auto-triggering domain-restricted web search (NDMA/ASDMA/CWC/IMD only) when local context is stale or insufficient.
 - Indexed **12,000+ chunks** from 27 OCR-processed government PDFs via ChromaDB, and built a live flood-risk map with a composite index blending real-time rainfall and river-discharge data across 9 gauge stations.
 - Technologies:** Python, LangChain, Gemini API, ChromaDB, Streamlit, Folium.
- MediAssist AI - Multimodal Clinical RAG Assistant [Demo](#) / [GitHub](#)
 - Built a multimodal clinical RAG assistant grounded in WHO/MedlinePlus guidelines (**9,600+ chunks**), combining **Gemini Vision** for wound/lab-report image analysis with bilingual (Hindi/English) voice I/O and turn-aware LLM prescribing logic.
 - Implemented a **drug-interaction and allergy safety layer** with cross-reactivity detection (e.g., penicillin allergy flags all β -lactams), generating downloadable clinical prescriptions via ReportLab.
 - Designed **bcrypt-based PIN authentication** with device-ID data isolation for secure, privacy-conscious patient access on shared devices.
 - Technologies:** Python, Gemini API, LangChain, ChromaDB, Streamlit, SQLite.

EXPERIENCE & LEADERSHIP

ANTERIX Club, VIT Bhopal – Student Developer	Nov'24 – Mar'25
Built and deployed the official club website and event management system, streamlining registrations and boosting engagement.	
ANTERIX Club, VIT Bhopal – President	Jan'25 – Sept'25
Led a 60+ member team , driving inclusive leadership and club-wide engagement initiatives.	

PUBLICATIONS

- IEEE Research Paper: *Network Coding for Fault-Tolerant Transmission of Biomedical Data* — Published in IEEE Xplore (IC3 2025); demonstrated **20% higher fault tolerance** in biomedical data transmission. (Oct'24 – Feb'25) — [Read](#)

ACHIEVEMENTS

- Ranked Top 40/300+ teams (1200+ participants) at Dawn of Code (Unigoal & Bhopal Municipal Corporation) by building SpecTrum, an AI-powered LMS with AWS S3 backend, boosting productivity by 40%.
- Recognized by the State Government for outstanding academic performance in Class 12 (CBSE), receiving the Chief Minister's Award, Assam.